Research Assignment 1
Bright Learn - Data Analysis - SQL

## Section A: Data Fundamentals

① What is a Database? Define it in your own words and explain why databases are essential in the modern world? Provide 2 real-world examples of where databases are used.

* A database is a organised body of information or collection of facts. Information or collection of facts is called data and we use data to derive decisions and turn them into action. A database is like a digital filing system that stores information in tables, rows, and columns ~~and~~ what for refer to as structured Data.
* Databases are important in the modern word because they allow us to access and retreive information instantly in a query. They also allow us to store information in an orderly fashion, especially large volumes of data, and we can store data more accurately and consistently.
* Banks store consumer accounts and records. Online stores, social media platforms, hospitals etc.

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② What ~~are~~ is a data Warehouse? How does it differ from a regular data base? Why would a company use a data warehouse?

* Data warehouse is a centre place where we store multiple databases.
* It differs from a database because databases store operational current data & data warehouse stores historical, multiple data.

* Companies use data warehouses to centralize large volumes of data from multiple sources so they can analyze trends and run reports.

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③ What is data Lake? Explain the concept and describe how differs from a data warehouse. When would a business prefer a data lake over a data warehouse.

* A data Lake stores raw uprocessed data.
* It differs from a data warehouse because you can store structured or unstructured data.
* Businesses would opt for a data Lake over data warehouse in instances where very large volumes of diverse data have to be stored.

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④ What is a data Mart? How does it relate to a data warehouse and who typically uses it?

* Data Mart is filtered data
* Data Mart relationship to data warehouse is that data warehouses handle large scale data and act as a source and data Mart is more filted and purpose-built slice of that data.
* Organizations, Executives & Managers

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⑤ Explain the difference between structured, Semi-structured & unstructured data, Provide one example of each.

* Structured data is stored in a table with rows and columns, it is consistent. Ustructured data is Least organized like social media posts, videos etc. Semi-structured data is between

the 2, such as XML documents.
Example of structured data is a sales record.

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⑥ What is data Modelling? Why is it important when designing a system that works with data? Describe atleast 2 types of data models

* Data modeling is a process that we do to define how data is structured. It is the identifying entity.
* Relational Database (RDB) & Database management system (DMS)

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⑦ What is data mining? How is it used in business. Provide two real-world examples of data mining in action.

* Data mining is when we analyse data to determine patterns by categorizing items or grouping them. This helps us to to determine factors and predict outcomes.
e.g: in retail like pick & pay or checkers, they use our transactions to determine patterns so they can determine product placing.

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⑧ What is Database Management Systems (DBMS)? Name and briefly describe four popular DBMS platforms (eg. MSQL, PostgreSQ Oracle, MongoDB) and explain a suitable use case for each.

* DBMS keeps data organized, safe & accessible.
My SQL is relational DBMS. it is structured query language.
Oracle is enterprise-level relational DBMS and is commonly used by banks.
Mongo DB is non-relational and is designed to handle unstructured/semi-structured data
Microsoft SQL is relational DBMS and is used for business inteligence

(9) What is ETL (Extract, Transform, Load)? Explain each step and describe why this process is important in data analytics.

* A process used to prepare data from various sources.
Extract: collecting data
Transform: clean up extracted data and structure it.
~~Transform~~ Load: after transforming load onto target systems
e.g (data warehouse, data mart)

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(10) What are common data storage formats used in data analyt (e.g CSV, JSON, Parquet, Avro)? Briefly describe each and state when you should use one over another.
CSV (commo-separated values): plain text where fields are separated by commas. (better to use for simple data)
JSON (JavaScript Object Notation): flexible format used in semi-structured data, better to use for big data
Parquet: columnar storage format meaning queries are stored by columns
Avro: binary format: shema (better to use for big data)

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(11) What is artificial intelligence? (AI) Define it and explain the difference between Narrow AI (Weak AI) and general AI (strong AI) Provide one example of each.

* AI is the field of computer science that's all about creating systems or machines that can perform tasks.
* Narrow AI is for specific tasks like face recognition
* General AI is used to perform intellectual tasks that a human can do. It is futuristic so they are actual examples just only in theory.

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(12) What is machine learning? (ML) How does it differ from traditional programming? Describe the main three types of machine learning: Superused, Unsupervised & Reinforcement Learning.

* ML is a branch of AI where computers can learn patterns from Data. Tradional programming needs a developer to write a clear set of rules.
* Supervised ML: give inputs & outputs, it is used for tasks like classifications e.g: spam detection.

Unsupervised ML: just inputs, model uses clustering

Reinforement ML: this model interacts with environments and learns strategies over time

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(13) What is deep learning? How does it relate to Machine Learning and Artificial Intelligence? What are Neural Networks and how do they work?

* Deep learning is a portion of Machine Learning, it relies on artificial neural networks. Neural Networks are a key component in Deep learing. They are made up of layers of interconnected nodes or neurons. Each layer takes in an input which is passes through an activated function and moves on to the next layer.

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